

AGCL: Aspect Graph Construction and Learning for Aspect-level Sentiment Classification

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Abstract

Prior studies on Aspect-level Sentiment Classification (ALSC) emphasize modeling inter-relationships among aspects and contexts but overlook the crucial role of aspects themselves as essential domain knowledge. To this end, we propose **AGCL**, a novel Aspect Graph Construction and Learning method, aimed at furnishing the model with finely tuned aspect information to bolster its task-understanding ability. AGCL’s pivotal innovations reside in Aspect Graph Construction (**AGC**) and Aspect Graph Learning (**AGL**), where AGC harnesses intrinsic aspect connections to construct the domain aspect graph, and then AGL iteratively updates the introduced aspect graph to enhance its domain expertise, making it more suitable for the ALSC task. Hence, this domain aspect graph can serve as a bridge connecting unseen aspects with seen aspects, thereby enhancing the model’s generalization capability. Experiment results on three widely used datasets demonstrate the significance of aspect information for ALSC and highlight AGL’s superiority in aspect learning, surpassing state-of-the-art baselines greatly. Code is available at <https://github.com/jian-projects/agcl>.

1 Introduction

As an important research topic of Natural Language Processing (NLP), Aspect-level Sentiment Classification (ALSC) is a fine-grained sentiment analysis task that aims to identify the sentiment polarity of a review text toward each corresponding aspect (Brauwers and Frasinicar, 2022; Jian et al.,

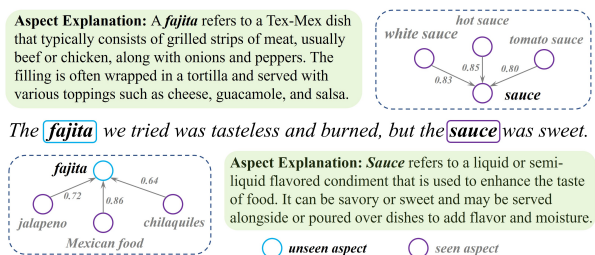


Figure 1: An example of review text. Seen aspects are that explicitly mentioned in the training set, while unseen aspects are not.

2024), which has been widely applied in real-world scenarios such as public opinion monitoring (Chen et al., 2022b), text classification (Bestvater and Monroe, 2023), and content recommendation (Kim et al., 2021). As illustrated in Figure 1, given the review text “The *fajita* we tried was tasteless and burned, but the *sauce* was sweet.”, ALSC is required to predict the sentiment polarities toward “*fajita*” as *Negative* and “*sauce*” as *Positive*.

The rise of deep learning and Pre-trained Language Models (PLMs) (Qiu et al., 2020) has significantly advanced NLP tasks with high accuracy (Min et al., 2023). Therefore, leveraging PLMs as the foundation, various ALSC methods have been developed to investigate inherent connections between aspects and contexts. Among them, RNN-based approaches (Wankhade et al., 2023; Huang et al., 2023) prioritize capturing surface-level word sequence information, while Graph-based models (Li et al., 2021a; Yang et al., 2023; Jiang et al., 2023; Yin and Zhong, 2024) exploit sentence dependency structures to account for syn-

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tactic relationships, often resulting in superior performance over RNN-based counterparts. In addition, Knowledge-based models (Li et al., 2021b; Zhang et al., 2022b; Yong et al., 2023; Jian et al., 2024; Ouyang et al., 2024) focus on improving the model’s understanding of domain-specific tasks by integrating external valuable information and have emerged as the predominant approaches within the ALSC community. Their success lies in the alignment of domain knowledge with the general knowledge (Bao et al., 2023). As shown in Figure 2, "general knowledge" refers to the knowledge stored within PLMs, while "domain knowledge" pertains to fine-tuning data. Additionally, we designate "aligned knowledge" as that contained within the ALSC model after fine-tuning with domain data.

Inspired by Jian et al. (2024), aspects are pre-defined specialized terms and phrases, often rare in general language usage particularly within low-resource language environments. They serve as prime representatives of domain knowledge within the ALSC scenario, prompting two pivotal issues: 1) prior studies have predominantly concentrated on modeling the aspect-opinion interplay, neglecting the crucial role of aspects as representatives of domain knowledge, and 2) aspects present in the training set, referred to as "seen aspects", undergo explicit alignment after fine-tuning with training data, while the majority of "unseen aspects" (those absent from the training data) remain under-aligned, hindering the model’s generalization ability. Although the subword-based tokenization technique, *e.g.*, BPE (Sennrich et al., 2016), used in PLMs alleviate the issue of under-alignment to some extent, the limited number of seen aspect tokens fails to adequately cover all subwords associated with unseen aspects.

To address these issues, we propose a novel Aspect Graph Construction and Learning (AGCL) method to highlight the important role of aspects for ALSC. Based on the interrelationships among domain aspects, we suggest building the domain aspect graph, which is leveraged as expert knowledge to facilitate both model training and inference processes. Figure 1 depicts a subset of nodes and edges within the built aspect graph, where nodes represent aspects and edges denote their similarities. Hence, this aspect graph can be used as a bridge to link the unseen aspect (*e.g.*, "fajita" in Figure 1) with seen aspects, and thus provides a way to deduce representations of unseen aspects through their associations with seen aspects.

Given the expenses of building the aspect graph by domain experts, we suggest treating language models as experts to automate the aspect graph construction, dubbed AGC. To bolster the efficacy of the aspect graph, we develop the Aspect Graph Learning (AGL), including two key iterative processes: 1) enhancing aspect representations with valuable information derived from the aspect graph, and 2) updating the aspect graph with aligned knowledge to enhance its domain expertise and efficacy. Moreover, based on the similarities of aspects, we introduce contrastive learning to pull similar aspects closer and push dissimilar aspects apart, thereby improving the robustness of aspect representations. In summary, our contributions are summarized as follows:

- We highly suggest building the domain aspect graph, utilized as expert knowledge, to enhance the ALSC model’s focus on aspects and improve its generalization capability.
- We carefully develop the aspect graph learning method to facilitate knowledge alignment and achieve a more refined and effective aspect graph for sentiment analysis.
- We extensively experiment on three ALSC datasets, yielding promising results that affirm the significance of aspect information for ALSC, and the effectiveness of AGL in aspect learning.

2 Related Work

2.1 Aspect-Level Sentiment Classification

Due to the strong language modeling capabilities, PLMs have become the primary choice for ALSC (Brauwers and Frasincar, 2022). Existing PLM-based ALSC methods broadly fall into two categories: those utilizing PLMs as text encoders and those enhancing PLMs’ comprehension abilities.

Building upon the PLMs, sophisticated model components are intricately crafted to discern the correlation between contextual opinions and aspects, mainly including attention mechanisms and graph structure learning. Representatively, Tang et al. (2019) and Su et al. (2021) iteratively masked tokens with the highest attention weights to uncover the most influential opinion words. In contrast, GNN-based methods (Wang et al., 2020; Xiao et al., 2021; Li et al., 2021a; Chen et al., 2022a; Ma et al., 2023; Yin and Zhong, 2024) usually

introduced the syntactic dependency tree knowledge and employed GNN to encode and analyze the structural relationships within the text, where DualGCN (Li et al., 2021a) designed SynGCN to alleviate dependency error and SemGCN to capture semantic correlations, dotGCN (Chen et al., 2022a) proposed an aspect-specific and language-agnostic discrete latent opinion tree structure to reduce the dependency on the accuracy of the parse tree, and APARN (Ma et al., 2023) replaced the syntactic dependency tree with the semantic structure to align the semantic requirement for the ALSC task.

Without complex components design, Li et al. (2019), Xu et al. (2019) and Silva and Marcacini (2021) demonstrate that remarkable results can be achieved by merely appending a linear classification layer and then fine-tuning PLMs with few domain data. This proficiency stems from aligning pre-trained general knowledge with domain-specific knowledge. Hence, numerous studies have attempted to incorporate external knowledge to enhance the model’s task-understanding ability. On the one hand, further exploiting the domain knowledge in the training set can significantly improve the model’s performance. Typically, Jian et al. (2024) proposed to retrieve similar samples from training data to execute joint learning, which enables the model to be aware of the unified pattern of sentiment semantics. On the other hand, external knowledge bases can bring additional information to enhance the available general and domain knowledge. For example, Zhong et al. (2023) introduced the knowledge graphs of WordNet (Miller, 1995) as prior knowledge to alleviate the difficulty of sentence comprehension. Wu et al. (2023) applied YAGO (Rebele et al., 2016) to extract additional entity information to mine the potential sentiment polarity of sentiment items. Jin et al. (2023) requested the Oxford Dictionary to expand the description of aspect terms.

In our work, we concentrate on the aspect information within the domain, emphasizing their importance for ALSC. The proposed AGL aims to enhance the model’s understanding by providing finely tuned aspect information, and establish a bridge to connect unseen aspects and seen aspects, thereby improving the model’s generalizability.

2.2 Contrastive Learning in ALSC

Contrastive learning (He et al., 2020; Gao et al., 2021; Xu et al., 2023) has emerged as a powerful paradigm in the domains of unsupervised represen-

tation learning (Gidaris et al., 2018) and supervised representation learning (Khosla et al., 2020). This approach leverages the notion that semantically similar samples should be brought closer in the embedding space while pushing dissimilar samples apart (Xu et al., 2023). Contrastive learning has been extensively applied in the ALSC task and has shown promising results. For example, Liang et al. (2021) utilized a supervised contrastive learning framework to exploit correlations and variances in sentiment polarities and patterns. Jian et al. (2024) proposed a retrieval contrastive learning method to enhance the model’s ability to capture the robust sentiment semantics of aspects. Shi et al. (2024) designed a KL divergence-based contrastive learning that promotes contextual representation modeling by incorporating dual-way information.

3 Methodology

3.1 Problem Definition and Motivation

Given a review text $T = \{t_1, t_2, \dots, t_n\}$ of n tokens with k aspects $\{a_i\}_{i=1}^k$, where each aspect is explicitly mentioned in T and spans across m_i ($1 \leq m_i < n$) tokens. ALSC aims to predict the sentiment polarity of T toward each aspect a_i , formulated as $f_{ALSC} : \mathcal{M}(T, a_i) \rightarrow \hat{y}_i$, where $\mathcal{M}$ is the PLM-based ALSC model, generally consisting of an encoder and classifier, that maps the input text to the sentiment polarity $\hat{y}_i \in \{Positive, Neutral, Negative\}$.

Given a domain data $D = \{\langle T_i, a_i, y_i \rangle\}_{i=1}^N$ with N samples, each comprised of a review text, a certain aspect, and its corresponding sentiment polarity. All aspects in D , denoted as A , construct the set of seen aspects, whose tokens can be explicitly trained to refine their semantics to meet the task’s requirement. In contrast to seen aspects, the unseen aspect $a \notin A$ cannot be explicitly trained, and its semantics heavily rely on the general knowledge of PLMs, leading to a disparity between domain-specific requirements and general knowledge. Hence, we suggest building the domain aspect graph and employing it as expert knowledge to establish a bridge between unseen aspects and seen aspects, thus mitigating the under-alignment issue of unseen aspects. In practice, the aspect graph is typically provided by domain experts, but the high cost of manual construction motivates us to develop an automated method for aspect graph construction, ensuring technical integrity.

3.2 Aspect Graph Construction (AGC)

Intuitively, aspects within a domain are relevant and can be connected by their intrinsic connections. Here, we leverage semantic similarities of seen aspects to construct the aspect graph. Unseen aspects can be inserted into this aspect graph based on their similarities to seen aspects, facilitating the semantics inference of unseen aspects from seen ones. Formally, we define the aspect graph as $\mathcal{G} = \{\mathcal{N}, \mathcal{E}\}$, where $\mathcal{N}$ is the set of nodes representing seen aspects, and $\mathcal{E}$ is the set of edges representing similarities of aspects. The challenging of aspect graph construction lies in determining values in $\mathcal{E}$. Here, we suggest treating language models as domain experts to encode the node representations and obtaining $\mathcal{E}$ automatically by cosine similarity calculation. Hence, the quality of $\mathcal{G}$ heavily relies on the abilities of language models.

Generally, aspects are words or phrases with few tokens, posing challenges for language models, even Large Language Models (LLMs) (Min et al., 2023), to accurately capture aspect semantics, thus resulting in imperfect aspect relationships. Inspired by the successful practice of LLMs in instruction learning (Ouyang et al., 2022), we request LLMs to elucidate aspects according to their corresponding domains. Subsequently, the Sentence Language Model (SLM), such as SBERT (Reimers and Gurevych, 2019), is employed to encode the aspect description and obtain aspect representation.

$$e_a = SLM(LLM(\text{aspect}, \text{domain})) \quad (1)$$

where LLM represents the process of leveraging LLMs to clarify the aspect term based on its domain. In this paper, the template of the prompt is designed as "You are a linguist in the domain of [domain], please succinctly explain what [aspect] means.". Two examples of aspect explanations are shown in Figure 1. SLM denotes the encoding process by a sentence language model and returns sentence embedding e_a as the representation of the corresponding aspect term.

In this way, informative aspect representations within the specified domain can be calculated, forming the node embedding attribution, denoted as $\mathcal{N}^e = \{e_{a_i}\}_{a_i \in A}$. Subsequently, similarities between aspects can be calculated:

$$\mathcal{E}(a_i, a_j) = \frac{e_{a_i} \cdot e_{a_j}}{\|e_{a_i}\| \|e_{a_j}\|} \quad (2)$$

where $a_i \in A$ and $a_j \in A$ are any two aspects in the training set. In addition, for the unseen aspect

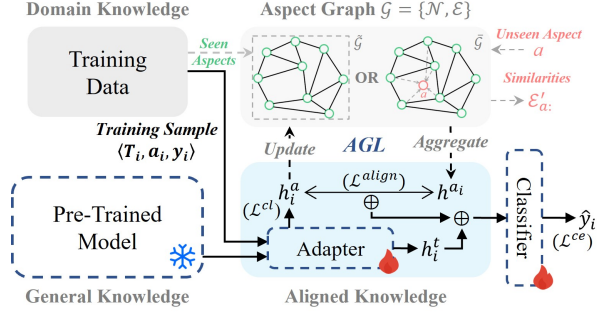


Figure 2: The process of Aspect Graph Learning (AGL), where the aspect graph can be built by AGC ($\mathcal{G} = \tilde{\mathcal{G}}$) or provided by domain experts ($\mathcal{G} = \tilde{\mathcal{G}}$). $\bullet$ indicates parameter unfrozen, while $*$ indicates parameter frozen.

$a \notin A$, its similarity with seen aspect $a_j \in A$ can be calculated as $\mathcal{E}'(a, a_j) = \frac{e_a \cdot e_{a_j}}{\|e_a\| \|e_{a_j}\|}$.

3.3 Aspect Graph Learning (AGL)

Fine-tuning the model with domain data is a common practice to enhance the model's understanding in the specified domain, a process of aligning general knowledge with domain knowledge, achieving aligned knowledge that is beneficial for the downstream task. In this process, representations of aspects will be constantly refined to adapt to the specified domain. Following the basic mode in Li et al. (2019) and Silva and Marcacini (2021), for each training sample $\langle T_i, a_i, y_i \rangle \in D$, token representations are calculated:

$$H = \mathcal{M}_e([CLS] T_i [SEP] a_i [SEP]) \quad (3)$$

where $\mathcal{M}_e$ is the model encoder used to transfer tokens into representations H . $[CLS]$ and $[SEP]$ are special tokens in the pre-trained model, where the token representation of $[CLS]$ is usually viewed as the overall text representation $h_i^t = H_0$. In addition, we employ mean pooling operation to aggregate multiple aspect token representations as the model-generated aspect representation $h_i^a \in \mathbb{R}^d$, where d is the dimension of representation.

3.3.1 Enhance Aspect Representation

As illustrated in Figure 2, in addition to the model-generated aspect representation, another kind of aspect representation can be derived from the aspect graph by aggregating other aspect representations based on their similarities:

$$\begin{cases} h^{a_i} = \sum_{j=1}^{|A|} w_j \cdot \mathcal{N}^e(a_j) \\ w = \text{norm}(\{\mathcal{E}(a_i, a_j)\}_{j=1}^{|A|}) \end{cases} \quad (4)$$

where h^{a_i} denotes the aggregated aspect representation obtained by the weighted sum of other aspect representations. $\mathcal{N}^e(*)$ represents retrieving the corresponding aspect representation from the domain aspect graph. $norm(*)$ denotes the normalization function that projects the similarities into weights. Thus, w_j denotes the proportion of j -th aspect representation in constructing h^{a_i} , satisfying $\sum_{j=1}^{|A|} w_j = 1$ and $w_i = 0$ (we exclude the target aspect itself to simulate the model’s inference scenario). For any unseen aspect $a \notin A$, its aggregated representation can be calculated in the same way based on its similarities to the seen aspects: $\mathcal{E}'_a = \{\mathcal{E}'(a, a_j)\}_{j=1}^{|A|}$.

In the inference phase, aggregated aspect representations are essential for unseen aspects, as we initially suppose their representations generated by the model are unconvincing. Hence, to ensure the quality of the aggregated aspect representation, we introduce the aspect representation alignment loss:

$$\mathcal{L}_i^{align} = \frac{1}{d} \sum_{j=1}^d |h_{i,j}^a - h_j^{a_i}| \quad (5)$$

where $h_{i,j}^a$ and $h_j^{a_i}$ are the j -th dimension of the model-generated aspect representation and the aggregated aspect representation, respectively. In this way, the aggregated aspect representations are encouraged to be close to the model-generated aspect representations to ensure their validity.

Finally, the enhanced aspect representation is obtained by combining the model-generated and the aggregated aspect representations:

$$\tilde{h}_i^a = (1 - \lambda)h_i^a + \lambda h^{a_i} \quad (6)$$

where $\lambda = 0.5$ is a weight coefficient that controls the influence of aggregated aspect representation. Ideally, for the seen aspect, the aggregated aspect representation matches the model-generated representation perfectly, and thus $\tilde{h}_i^a = h_i^a$. For the unseen aspect, the model-generated representation is effectively enhanced by the convincing aggregated aspect representation, thereby improving the model’s generalization ability.

3.3.2 Update Aspect Graph

The most important element in $\mathcal{G} = \{\mathcal{N}, \mathcal{E}\}$ is the aspect similarity matrix $\mathcal{E}$, which determines the relationships between aspects. As the inferior important element, we initialize the node attributes $\mathcal{N}^e$ with the aspect representations calculated by

SLMs and update them with the model-generated aspect representations:

$$\mathcal{N}^e(a_i) = \alpha_i \mathcal{N}^e(a_i) + (1 - \alpha_i)h_i^a \quad (7)$$

where $\alpha_i \in [0, 1)$ is a momentum coefficient (He et al., 2020) for node embedding updating. We employ the reciprocal of sample frequency as α_i : $\alpha_i = 1/N_{a_i}$, N_{a_i} is the number of samples with the aspect of a_i . This setting guarantees that comprehensive aspect knowledge from the training data can be incorporated into $\mathcal{N}^e$ within one epoch, irrespective of the aspect’s frequency. Furthermore, by updating $\mathcal{N}^e$ with aspect representations generated by the model, the aspect graph undergoes continual refinement tailored to domain-specific requirements, thereby bridging the gap between model-generated and aggregated aspect representations, *i.e.*, aggregated aspect representations can be viewed as aligned knowledge.

3.3.3 Aspect Representation Rectification

During the model training, we update node representations $\mathcal{N}^e$ with the model-generated aspect representations, which may lead to inconsistencies between the relationships reflected in $\mathcal{N}^e$ and those depicted in $\mathcal{E}$. One approach is to implement a hard constraint to ensure that the aspect representations comply with their similarities in $\mathcal{E}$. However, given potential errors in the automatically built aspect graph, we opt for soft constraints to regularize the model-generated aspect representations, *i.e.*, based on the aspect similarities within $\mathcal{E}$, we expect the representations of similar aspects to be close to each other and dissimilar aspects to be far apart. More precisely, for aspect representations within a *batch*, we retrieve their similarities from $\mathcal{E}$ and utilize contrastive learning to pull similar aspects together while pushing apart dissimilar ones.

$$\mathcal{L}^{cl} = -\frac{1}{N_b^i} \sum_{i=1}^{N_b} \mathbb{I}_{N_b^j > 0} \frac{1}{N_b^j} \sum_{j=1}^{N_b} \mathbb{I}_{\mathcal{E}_{ij} > \epsilon} \cdot \log \frac{\exp(\text{sim}(h_i^a, h_j^a)/\tau)}{\sum_{k=1}^{N_b} \mathbb{I}_{i \neq k} \cdot \exp(\text{sim}(h_i^a, h_k^a)/\tau)} \quad (8)$$

where h_i^a and h_j^a are the aspect representations of the i -th and j -th samples respectively, and $\mathcal{E}_{i,j}$ denotes their similarity that derived from the aspect graph. h_k^a is the aspect representation of the k -th sample in this *batch*. $\tau = 1.0$ is the temperature coefficient used for the cosine similarity measure function $\text{sim}(*, *)$. $\mathbb{I}_{condition}$ denotes the indicator

function that returns 1 when *condition* is satisfied, and 0 otherwise. N_b denotes the sample number within this batch, N_b^j and N_b^i indicates the numbers that meet the conditions $\mathcal{E}_{i,j} > \varepsilon$ and $N_b^j > 0$, respectively. $\varepsilon = 0.9$ is the threshold used to judge whether aspects are similar or not.

3.4 Model Training and Inference

Training: To further highlight the significance of the aspect, the text and aspect representations are joined to predict the sentiment polarity.

$$\hat{y}_i = \mathcal{M}_c(h_i^t + \tilde{h}_i^a) \quad (9)$$

where $\mathcal{M}_c$ denotes the classifier used to obtain the sentiment distribution $\hat{y}_i$. Then, the classification loss is calculated by the cross-entropy function:

$$\mathcal{L}_i^{ce} = -y_i \cdot \log(\hat{y}_i) \quad (10)$$

where y_i and $\hat{y}_i$ denote the ground truth and predicted sentiment distribution, respectively. Finally, the model is optimized by the combined loss:

$$\mathcal{L} = \frac{1}{N_b} \sum_{i=1}^{N_b} (\mathcal{L}_i^{ce} + \mathcal{L}_i^{align}) + \mathcal{L}^{cl} \quad (11)$$

Inference: As depicted in Table 1, aspects in the test set contain both seen and unseen aspects. For the sample with the seen aspect, we utilize the same mode as training to infer its sentiment polarity. For the sample with unseen aspect, the difference is that we need to additionally calculate the similarities between the unseen aspect and seen aspects due to they are not contained in the aspect graph.

4 Experimental setup

4.1 Datasets

We evaluate the proposed method on three widely-used ALSC datasets, including Laptops and Restaurants from SemEval 2014 Task 4 (Pontiki et al., 2014), and Twitter from Dong et al. (2014). The statistic of each dataset is summarized in Table 1, demonstrating that unseen aspects are prevalent in ALSC scenarios.

4.2 Compared Models

We compare AGL with recent advanced models, broadly divided into two categories: structure-based models and knowledge-based models.

Structure-based Models: BERTABSA-ATT (Su et al., 2021), DualGCN (Li et al., 2021a), dot-GCN (Chen et al., 2022a), BiSyn-GAT (Liang

Datasets	Laptops		Restaurants		Twitter	
	Train	Test	Train	Test	Train	Test
Aspects:						
<i>seen</i>	949	154	1202	187	113	77
<i>unseen</i>	-	235	-	333	-	5
Samples:						
<i>Positive</i>	994	341	2164	727	1561	173
<i>Neutral</i>	464	169	637	196	3127	346
<i>Negative</i>	870	128	807	196	1560	173
<i>Total</i>	2328	638	3608	1119	6248	692

Table 1: Statistics of the three ALSC datasets.

et al., 2022), RoBERTa4GCN (Xiao et al., 2021), TextGT+BERT (Yin and Zhong, 2024).

Knowledge-based Models: BERTABSA (Su et al., 2021), ABSA-DeBERTa (Silva and Marcacini, 2021), ABSA-ESA (Ouyang et al., 2024), DR-BERT (Zhang et al., 2022a), DeBERTa+RCL (Jian et al., 2024), PConvRoBERTa (Feng et al., 2023).

4.3 Implementation Details

For AGC, we utilize the GPT-3.5-turbo to generate aspect descriptions, and employ the SBERT model (specifically all-roberta-large-v1¹) to encode these enriched aspect descriptions with 1024 dimensions. We randomly sampled 100 aspect descriptions from each dataset for manual evaluation, and all were rated as acceptable, as exemplified in the Appendix.

For AGL, we adapt the DeBERTa-large² model with adapters embedded in each layer as the model encoder $\mathcal{M}_e$. The parameters of the pre-trained DeBERTa model are fixed, and only the parameters of adapters and classifier are fine-tuned during the model training. Hence, only 6.03% (26M/431M) of the parameters are fine-tuned, which significantly reduces the training cost. During the model training, Adam is utilized as the optimizer with the initial learning rate tuned from 1e-4 to 3e-4, the batch size is manually adjusted from 16 to 32, and the dropout rate is set to 0.3. The max number of epochs is set to 25, 25, and 30 for Laptops, Restaurants, and Twitter, respectively. The other hyperparameters have been provided when they are introduced in the Methodology. All experiments are conducted on a single NVIDIA 3090ti GPU with 24GB memory.

¹<https://huggingface.co/sentence-transformers/all-roberta-large-v1>

²<https://huggingface.co/microsoft/deberta-large>

Categories	Models	Laptops		Restaurants		Twitter	
		Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy
Prompt	GPT-3.5-turbo	73.16	80.00	75.17	87.15	61.67	60.96
Structure	DualGCN	78.10	81.80	81.16	87.13	76.02	77.40
	dotGCN	78.10	81.03	80.49	86.16	77.00	78.11
	BiSyn-GAT	79.15	82.44	81.63	87.49	76.80	77.99
	RoBERTa4GCN	78.16	81.80	78.61	86.23	74.00	74.75
	TextGT+BERT	78.71	81.33	82.27	87.31	76.45	77.70
Knowledge	♣BERTABSA	77.88	81.38	80.49	86.61	75.67	76.59
	BERTABSA-ATT	79.30	82.64	82.34	87.86	76.45	77.60
	ABSA-DeBERTa	79.36	82.76	83.42	<u>89.46</u>	-	-
	ABSA-ESA	79.34	82.44	81.74	<u>88.29</u>	-	-
	DR-BERT	78.16	81.45	82.31	87.72	76.10	77.24
	DeBERTa+RCL	80.28	82.76	<u>84.68</u>	89.38	77.47	78.32
	PConvRoBERTa	<u>80.89</u>	<u>83.54</u>	84.27	89.29	<u>77.53</u>	<u>78.47</u>
	AGL (Ours)	82.15 _{↑1.26}	84.54 _{↑1.00}	85.63 _{↑0.95}	90.30 _{↑0.84}	78.15 _{↑0.62}	78.85 _{↑0.38}

Table 2: Comparisons (%) among baselines, with best and second-best results highlighted in bold and underlined, respectively. ♣ denotes the results are derived from Su et al. (2021), others are cited from their original publications.

5 Experimental results

5.1 Main Results

We run our model three times and compare it with advanced baselines. The main comparative results are tabulated in Table 2, with the best and second-best results highlighted in bold and underlined, respectively. We additionally using the GPT-3.5-turbo model to execute the ALSC task (Zhang et al., 2024) (refer to the Appendix for more details). Despite LLM’s superior performance on universal tasks, it lags behind the specialized models for the ALSC task, indicating the necessity of designing specialized models for specific tasks. In addition, we have the following observations.

First, PLM-based models excel well in ALSC because PLMs have learned a large amount of general language knowledge from extensive corpora, facilitating the capture of intricate syntactic and semantic nuances. As evidenced by BERTABSA, simply fine-tuning the BERT model yields competitive results. Furthermore, incorporating external dependency syntax tree knowledge is effective, approaches like DualGCN and BiSyn-GAT surpass BERTABSA across all datasets. The difference between structure-based models lies in the usage mode of the dependency syntax tree, and they possess their strengths and weaknesses in different scenarios. For example, dotGCN outperforms the other structure-based models on Twitter but lags on Laptops. TextGT+BERT performs well in Restau-

rants but falls short on Laptops and Twitter, inferior to BiSyn-GAT. The potential limitation may lie in the variability of parsed dependency syntax trees across different datasets, which could affect the model’s ability to generalize effectively.

Second, knowledge-based models concentrate on enhancing the model’s understanding of the data and task attributes. BERTABSA-ATT leverages the most influential tokens within the sentence, resulting in significantly improved performance compared to BERTABSA. Compared with ABSA-DeBERTa, DeBERTa+RCL achieves better performance through the retrieval of similar samples and joint training with these retrieved samples. PConvRoBERTa and DeBERTa+RCL surpass the structure-based models across all datasets, indicating the critical importance of enhancing the model’s comprehension of data and task attributes in the PLM era. The potential advantage of knowledge-based models may be that they further activate the PLM’s ability to capture the intrinsic sentiment semantic of the review text, which is more conducive to the model’s generalization.

Finally, AGL surpasses all comparative baselines in both Accuracy and Macro-F1 scores. Compared to the second-best results, AGL achieves improvements of 1.26%, 0.95%, and 0.62% on Laptops, Restaurants, and Twitter, respectively, in terms of Macro-F1. In addition, AGL outperforms ABSA-DeBERTa by enhancing semantics of aspects, re-

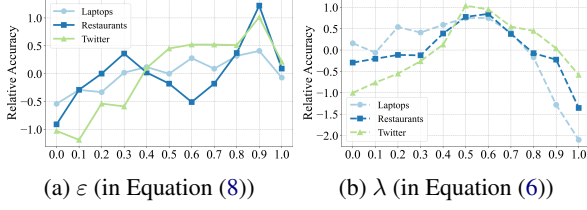


Figure 3: AGL’s performance with different parameters.

Methods	<i>seen</i>	<i>unseen</i>
w/ AGL	91.89	86.84
w/o AGL	90.93 $\downarrow$ 0.96	85.42 $\downarrow$ 1.42

Table 3: Accuracies of samples with different aspects.

sulting in significant performance enhancements of 2.79 points on Laptops and 2.21 points on Restaurants in terms of the Macro-F1 metric. These results underscore the significance of aspect information for ALSC and the effectiveness of AGL in learning meaningful aspect semantics.

5.2 Parameter Analysis

As summarized in Figure 3, we investigate the impacts of hyperparameters ε (threshold of similarity in Equation (8)) and λ (weight coefficient in Equation (6)) on the model’s performance (both vary from 0.0 to 1.0). Relative accuracy (values adjusted by subtracting the mean values) is adapted to facilitate the comparison. From Figure 3(a), AGl’s performance is gradually improved with the increase of ε , and reaching the optimal value around 0.9. $\varepsilon = 1.0$ means that the aspect representation rectification module is disabled, which leads to a significant performance drop. Additionally, Figure 3(b) shows that introducing aspect representations from the optimized aspect graph improves the model’s performance, achieving the best results at $\lambda = 0.5$.

5.3 Ablation Studies

In this section, we conduct ablation studies to identify the key factors behind AGl’s superiority.

Effects on different aspects: Table 3 statistics accuracies of samples with seen and unseen aspects on Restaurants. As seen, without using AGl, the model’s performance declines in both seen and unseen aspects, with a more pronounced drop in samples with unseen aspects. This result highlights the effectiveness of AGl in enhancing the semantics of aspects, particularly unseen aspects, which improves the model’s generalization ability.

Methods	Macro-F1	Accuracy
AGL	85.63	90.30
w/o $[\mathcal{L}^{cl}]$	84.50 $\downarrow$ 1.13	89.76 $\downarrow$ 0.54
w/o $[\mathcal{L}^{align}]$	85.33 $\downarrow$ 0.30	90.06 $\downarrow$ 0.24
w/o $[\mathcal{L}^{cl}, \mathcal{L}^{align}]$	84.17 $\downarrow$ 1.46	89.34 $\downarrow$ 0.96

Table 4: Ablation studies with different modules.

Influences of key modules: Two key modules, $\mathcal{L}^{cl}$ and $\mathcal{L}^{align}$, are removed to evaluate their impact on the model’s performance. Experimental results on Restaurants are presented in Table 4, from which we have the following observations. 1) Rectifying aspect representations is crucial, as removing $\mathcal{L}^{cl}$ causes a significant performance drop, highlighting their importance for aspect graph learning and knowledge alignment. 2) The primary role of $\mathcal{L}^{align}$ is to bridge unseen aspects, and its removal, resulting in a performance drop, highlights AGl’s ability to generalize to unseen aspects. 3) The performance drop is more pronounced when both $\mathcal{L}^{cl}$ and $\mathcal{L}^{align}$ are removed, highlighting the importance of both modules in AGl, they collectively contribute to AGl’s superiority.

5.4 Visualization of aspects

To further highlight the importance of our proposed AGl, we visualize aspect representations on Restaurants using t-SNE (Van der Maaten and Hinton, 2008) in Figure 4. Aspect representations are extracted from the model at a mid-performance checkpoint from three runs. As illustrated in Figure 4(a), due to the influence of the context, representations of the same aspects are scattered in the embedding space. After rectified by the aspect graph, aspect representations, for both seen (e.g., "price") and unseen aspects (e.g., "cake"), are kept consistent regardless of the context, as depicted in Figure 4(b). Furthermore, when AGl is introduced, similar aspects (e.g., "price" and "prices") are brought closer together while dissimilar ones (e.g., "price" and "cake") are pushed further apart, highlighting its notable superiority.

6 Conclusion

In this paper, we explore the role of aspect for ALSC instead of meticulous model component design. To this end, we propose AGCL, a novel method that constructs the domain aspect graph and employs it as expert knowledge to enhance the

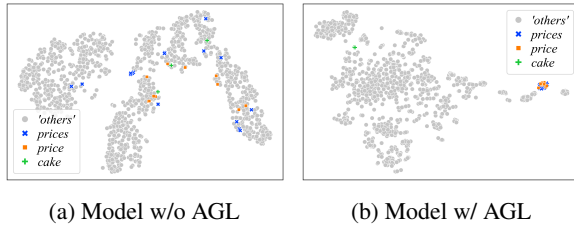


Figure 4: Visualization of aspects, "prices" and "price" are seen aspects, while "cake" is an unseen aspect.

aspect representations. The key points of AGCL reside in two perspectives: 1) employing language models as experts to automatically build the domain aspect graph, and 2) developing effective mechanisms to update the aspect graph and facilitate the knowledge alignment process. Experimental results and ablation studies demonstrate the effectiveness of AGL in enhancing the aspect representations and improving the model's generalization ability. In the future, we plan to explore better ways to use the well-refined aspect information in the domain aspect graph.

Limitations

This work has three limitations: 1) AGL depends on the domain expertise of aspect graph. Manual construction of this graph is costly, while automated construction relies on the performance of language models. 2) Information aggregation within the aspect graph is limited to one-hop, limiting deeper aspect dependency exploration. 3) This paper provides a straightforward implementation of aspect graph knowledge, while the more nuanced domain knowledge embedded in the constructed aspect graph remains under exploration and is reserved for future work.

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A Appendix

A.1 Aspect Descriptions Generated by GPT-3.5-turbo

To achieve better connections between aspects and their similarities, we use GPT-3.5-turbo to generate aspect descriptions with the template *"You are a linguist in the domain of [domain], please succinctly explain what [aspect] means."*. Some examples of the generated aspect descriptions are shown as follows. All the generated aspect descriptions are provided in the source code repository.

Generating aspect descriptions by GPT-3.5-turbo

Domain: Laptops **Aspect:** cord

Prompt: You are a linguist in the domain of Laptops, please succinctly explain what cord means.

Description: In the context of laptops, a cord usually refers to the power cord or charger that is used to supply power to the laptop and charge its battery.

Domain: Laptops **Aspect:** tech guy

Prompt: You are a linguist in the domain of Laptops, please succinctly explain what tech guy means.

Description: The term tech guy typically refers to someone who is knowledgeable and proficient in technology, particularly in the context of troubleshooting, fixing, or assisting with technical issues, often related to computers or other electronic devices. This person is often seen as the go-to individual for resolving technology-related problems.

Domain: Laptops **Aspect:** quality

Prompt: You are a linguist in the domain of Laptops, please succinctly explain what quality means.

Description: In the context of laptops, 'quality' refers to the overall standard and performance of the device, including factors such as durability, reliability, performance, design, and features. A high-quality laptop is one that meets or exceeds expectations in these areas.

Domain: Restaurants **Aspect:** staff

Prompt: You are a linguist in the domain of Restaurants, please succinctly explain what staff means.

Description: In the context of restaurants, staff refers to the group of people who work at the restaurant, including servers, chefs, managers, and other support or administrative personnel.

Domain: Restaurants **Aspect:** kitchen

Prompt: You are a linguist in the domain of Restaurants, please succinctly explain what kitchen means.

Description: The term 'kitchen' refers to a room or area where food is prepared and cooked in a restaurant or home.

Domain: Restaurants **Aspect:** menu

Prompt: You are a linguist in the domain of Restaurants, please succinctly explain what menu means.

Description: The term menu refers to a list of food and beverage items available for consumption at a restaurant or an eating establishment.

Domain: Twitter **Aspect:** jimmy carter

Prompt: You are a linguist in the domain of Restaurants, please succinctly explain what jimmy carter means.

Description: 'Jimmy Carter' refers to the 39th President of the United States, James Earl Carter Jr., who served from 1977 to 1981.

Domain: Twitter **Aspect:** hilary swank

Prompt: You are a linguist in the domain of Restaurants, please succinctly explain what hilary swank means.

Description: Hilary Swank is an actress known for her roles in movies such as Million Dollar Baby and Boys Don't Cry.

Domain: Twitter **Aspect:** google

Prompt: You are a linguist in the domain of Restaurants, please succinctly explain what google means.

Description: 'Google' is a verb that means to search for something on the internet using the Google search engine.

A.2 Performance of LLMs on Aspect-level Sentiment Classification

Following Zhang et al. (2024), we evaluate LLM performance on aspect-level sentiment classification (ALSC) using the SemEval 2014 datasets for laptops and restaurants, as well as the Twitter dataset. The prompts for zero-shot and few-shot settings are exemplified below, including the task definition, demonstrations, and input sentence. In few-shot settings, demonstrations are selected based on the similarity between input sentences and training sentences. We retrieve an equal number of samples across different sentiment polarities, thus 3, 6, and 9 demonstrations are used in our experiments, respectively.

Experimental results across different settings and models are shown in Table 5, where we employ more powerful GPT models to fully explore the potential of LLMs in the ALSC task. It’s worth noting that advanced models like GPT-4o are unnecessary for the aspect description generation in our work, as simple and concise aspect descriptions that we need can be effectively generated by GPT-3.5-turbo. Hence, we only report the best results of GPT-3.5-turbo in Table 2. As seen, more powerful models may not always lead to better performance, such as GPT-4o-mini performs better than GPT-4o on Laptops and Twitter. Furthermore, demonstrations are crucial for the model’s performance, which significantly improves the model’s performance in the few-shot setting. However, more demonstrations do not always lead to better performance, as better results can be achieved with 6 demonstrations than 9 demonstrations on most cases.

Models	Settings	Laptops		Restaurants		Twitter	
		Macro-F1	Accuracy	Macro-F1	Accuracy	Macro-F1	Accuracy
GPT-3.5-turbo	0-shot	57.21	73.22	63.78	81.56	50.35	51.98
	3-shot	70.87	79.04	73.00	85.64	60.31	59.77
	6-shot	73.16	80.00	72.30	84.62	61.67	60.96
	9-shot	71.22	77.60	75.17	87.15	61.06	60.64
GPT-4o-mini	0-shot	74.56	80.35	65.76	83.61	64.78	64.57
	3-shot	72.40	79.43	70.57	84.88	65.05	64.48
	6-shot	73.60	79.67	74.09	85.71	66.21	65.57
	9-shot	76.30	80.66	73.40	86.11	64.35	63.48
GPT-4o	0-shot	71.79	75.84	70.18	84.36	52.49	51.65
	3-shot	70.84	75.84	78.15	87.93	57.84	56.82
	6-shot	73.93	78.61	85.65	90.50	64.96	64.20
	9-shot	71.49	75.98	79.66	86.59	62.22	61.41

Table 5: Experimental results of LLMs on aspect-level sentiment classification.

Prompt for ALSC: zero-shot prompting

Definition:

Please perform the Aspect Level Sentiment Classification task:

given a sentence and a specific aspect, predict the sentiment of this sentence toward this aspect. Sentiment must be selected from ['negative', 'neutral', 'positive']. Please return the predicted sentiment only, without any other comments or texts.

Demonstrations:

Input:

Now, complete the task:

Sentence: **the bread is top notch as well .**

Aspect: **bread**

Output:

Label: **positive**

Prompt for ALSC: few-shot prompting (3-shot for example)

Definition:

Please perform the Aspect Level Sentiment Classification task:

given a sentence and a specific aspect, predict the sentiment of this sentence toward this aspect. Sentiment must be selected from ['negative', 'neutral', 'positive']. Please return the predicted sentiment only, without any other comments or texts.

Demonstrations:

Sentence: very good breads as well .

Aspect: breads

Label: positive

Sentence: the bread is the soft paratha bread (unlike the plain bread they use in calcutta) , and the stuffing is tandoori styled and very flavorful .

Aspect: bread

Label: negative

Sentence: also , top the meal with a delicious and perfect slice of tiramisu .

Aspect: meal

Label: neutral

Input:

Now, complete the task:

Sentence: **the bread is top notch as well .**

Aspect: **bread**

Output:

Label: **positive**